

Vismay Patel

📄 <https://vismay93.github.io/>

Summary

With **7.5 years of experience** as an Applied Scientist and Researcher, I specialize in **architecting, leading the development, and deploying scalable machine learning solutions** across **Computer Vision, Natural Language Processing, Deep Learning, and Generative AI**. My expertise includes building end-to-end systems for real-world applications such as **Multi-modal LLMs, Open-world Object Discovery, Document Understanding, Generative Image Models for redacted document correction, and Real-time Content Moderation**. Holding a Master's in Research from IIT Madras with a focus on Computer Vision and Deep Learning, I am passionate about translating cutting-edge research into production-ready, impactful solutions. I have a proven track record of technical leadership, mentorship of junior scientists and driving projects from concept to deployment at scale.

Experience

Applied Scientist

January 2022 - Present

Ads Trust Team, Amazon INC, Bangalore, India

- **Led the architecture and deployment** of a scalable and efficient multimodal content labeling system using cutting-edge VLMs (QwenVL, Ldefics2) with defect and marketplace-specific LoRA adapters, processing Images, Videos, and Product Pages achieving **90% label coverage at 95% precision**. (Technologies: SLoRA, VLLM, Sagemaker).
- Architected a regulatory compliance system for pharmaceutical ads to detect validate **Prescription (Rx) status** that works for **Indirect Advertising** with a Human-in-the-loop workflow achieving **99% Recall and 95% Precision**. (Technologies: Bedrock, Sagemaker, Lambda, Step Functions).
- Designed and implemented a real-time, interpretable content moderation system at scale, leveraging LLM experts to generate keyword-based decision stumps. (Technologies: Bedrock, Glue, Lambda, Step Functions).
- Developed and deployed small-scale multi-modal models (Flava, ConvNextV2) using Noise robust Semi-supervised training techniques for low-cost first-pass moderation.
- **Architected and built** a production-grade object-centric image retrieval platform capable of billion-scale fine-grained patch search. (Technologies: DINOv2, Edgeboxes, HNSW Index, Lambda, Step Functions).
- Spearheaded the development of an end-to-end synthetic data generation pipeline for AI-generated content moderation, combining image2prompt techniques with fine-tuned Latent Diffusion models. (Technologies: Diffusion Models).
- Implemented automated advertisement defect fixes using **Generative Image Models** for tasks like Inpainting, Object Detection, and Conditional Image Generation.

Machine Learning Engineer

April 2020 - January 2022

Global R&D Team, Verisk, Hyderabad, India

- **Architected and developed** an enterprise-scale document information extraction system leveraging active learning and document LMs. (Technologies: Active Learning, Layout Detection, Table Understanding, Key-Value extraction).
- Implemented a synthetic medical document generation framework combining layout generators with controlled text generation, producing HIPAA-compliant training data. (Technologies: Layout Generation, In-painting, MLM).
- Architected a large-scale pandemic modeling system using agent-based network simulations. Optimized computation using Cupy and custom graph diffusion algorithms.

Software Development Engineer

August 2018 - April 2020

Codenation LLC, Bangalore, India

- Deployed ML Factory, a production-grade AutoML platform enabling end-to-end model development for structured data. (Technologies: AWS Sagemaker, Django).
- Developed AutoFix: an innovative code quality improvement system leveraging AST graphs for automated code smell detection and remediation. (Technologies: Neo4j, Spring boot, AWS).

Internship

February 2018 - April 2018

Find Me A Shoe, Chennai, India

- Developed an algorithm for precise foot measurement extraction from multi-view images using camera geometry.

Publications

- **Vismay Patel**, and Shreya M. "ProDi-gy: Prompt-guided Distillation of LLMs for Fast and Interpretable Text Classification." In ACVC 2024 (Internal Conference).
- **Vismay Patel**, Mayank Gupta, and Pooja A. "Large-scale Patch-based Image Retrieval." In AMLC 2022 (Internal Conference).
- **Vismay Patel**, Niranjan Mujumdar, Prashanth Balasubramanian, Smit Marvaniya, and Anurag Mittal. "Data Augmentation using Part Analysis for Shape Classification." In WACV 2019.
- **Vismay Patel**, Arulkumar Subramaniam, Ashish Mishra, Prashanth Balasubramanian, and Anurag Mittal. "Bi-modal first impressions recognition using temporally ordered deep audio and stochastic visual features." In ECCV 2016 Workshops.
- **Vismay Patel**, Anubha Pandey, "Generative Image Inpainting for Person Pose Generation" and "Joint Caption Detection and Inpainting using Generative Network" published at ECCV 2018 Workshops.

Academic Projects

Data Augmentation using Part-based Deformations of Shapes

Aug 2015 - Feb 2018

Guide: Prof. Anurag Mittal

M.S. Project, Team Size:1

- Thesis research work focusing on learning image classifiers with limited labeled data.
- Proposed a method using part analysis of shapes to augment labeled data.
- Showed improved performance with CNN classifiers for shape classification using this augmentation. (Technologies: Matlab, Torch).

Human Pose Transfer

Oct 2017 - Nov 2017

Faculty: Dr. Mitesh Khapra, Topics in DL

Course Project, Team Size:2

- Used Variational Autoencoders (VAEs) to transfer the pose from one image to the human in another image. (Technology: Pytorch).

Technical Skills

Languages: Python, Java, Lua.

AI/ML Domains: Computer Vision, Natural Language Processing, Deep Learning, **Generative AI**, Multimodal Models, Explainable AI, Active Learning, Semi-supervised Learning.

ML Frameworks: PyTorch, Transformers, Sklearn, JAX, Tensorflow, OpenCV.

Generative Models: LLMs, VLMs (QwenVL, Idefics2), Latent Diffusion Models, GANs, VAEs.

ML Ops / Cloud: AWS, Kubernetes, Docker, Kubeflow, Sagemaker, VLLM, SLoRA, Bedrock, Glue, Lambda, Step Functions.

Databases / Other: Neo4j, HNSW Index, node-red.

Education

Indian Institute of Technology Madras

Chennai, India

M.S. (Computer Science and Engg.), **CGPA: 8.8/10**

2018

Specialization in Computer Vision and Deep Learning.

Institute of Technology, Nirma Uni.

Ahmedabad, India

B.Tech. (Information Technology), **CGPA: 7.76/10**

2014